

Soumya Subhra Datta

Artificial Intelligence Engineer

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PROFESSIONAL SUMMARY

Motivated Artificial Intelligence Engineer with hands-on experience in Python, Generative AI, Machine Learning, Deep Learning, and full-stack development. Skilled in building scalable AI systems, RAG pipelines, LSTM-based sequential models, and intelligent automation tools using OpenAI API, Gemini API, and LangChain. Proficient in prompt engineering, network intrusion detection, anomaly detection, and deploying real-world AI solutions using open-source LLMs and modern AI frameworks. Strong foundation in data science, computer vision, and end-to-end ML/DL pipeline development.

TECHNICAL SKILLS

Programming Languages:	Python, Java, JavaScript, C
Machine Learning & AI:	Scikit-learn, Random Forest, SGDClassifier, Naive Bayes, Anomaly Detection, Cost-Sensitive Learning, ROC-AUC, PR-AUC, Precision@K
Deep Learning:	LSTM, RNN, GRU, CNN, Transformer, BERT, Autoencoder, GAN, ResNet, VGG, MobileNet, Sequential Modeling, Time-Series Analysis, Network Traffic Classification, Transfer Learning, Attention Mechanisms
Data Science & Analytics:	Pandas, NumPy, Matplotlib, Tableau, Feature Engineering, Data Cleaning, Missing Value Handling, Outlier Removal, Class Imbalance Correction
Generative AI:	LangChain, OpenAI API, Gemini API, RAG Pipelines, Prompt Engineering, Memory Modules, FAISS, ChromaDB
NLP:	NLTK, SpaCy, Tokenization, Text Classification, Named Entity Recognition (NER), Sentiment Analysis, Word Embeddings, Word2Vec, GloVe, Text Preprocessing, Sequence Modeling
Computer Vision:	OpenCV, CNN Architectures, Image Preprocessing, Object Detection, Feature Extraction
APIs & Deployment:	REST APIs, Flask, FastAPI, Node.js, Hugging Face, Google Colab, Jupyter Notebook, Render
Frontend Development:	React.js, HTML, CSS, JavaScript
Databases:	MySQL, MongoDB Atlas, NoSQL
Tools & Platforms:	GitHub, SQL, Excel, dotenv, Cerberus API

PROJECTS

AI-Powered Proactive Threat Detection System — LSTM | Flask | MySQL | Real-Time

- Built a real-time cybersecurity web app using **Flask** and **MySQL** that monitors, classifies, and auto-mitigates threats via an **Identify-Grade-Mitigate** pipeline trained on **CICIDS 2017** (2.8M+ records). GitHub: github.com/Soumya-Subhra-Datta/AI-Threat-Prediction-System
- Trained an **LSTM** model across **4 telemetry features** achieving **93%+ accuracy**; implemented **4-class threat grading** with automated IP blocking via WAF/Nginx integration, reducing false positives by **25%**.

Apollo Metrics — AI-Powered Data Analyst Platform | Python | MySQL | LLM | Render

- Built a cloud-deployed data analysis platform on **Render** (live at: apollo-metrics.onrender.com) with a **Python** backend and **MySQL** database; users upload CSV datasets to trigger automated **EDA, ML model training, interactive visualization, and report generation**.
- Integrated an **LLM-powered chatbot** for natural language dataset queries and insights; supported one-click download of trained models, charts, and analytical reports via a responsive **HTML/CSS/JavaScript** frontend.

AI Drug Discovery Assistant — Flask | MySQL | Cerebras AI | Cheminformatics | SQLAlchemy

- Built a full-stack web application that predicts **molecular binding affinity, drug-likeness, toxicity risk, and solubility** from SMILES string inputs; integrated **Cerebras cloud AI** to generate natural-language pharmacological explanations and optimized candidate molecules. GitHub: github.com/Soumya-Subhra-Datta/AI-Based-Drug-Discovery_System
- Stored all molecular predictions and analysis history in **MySQL** via **SQLAlchemy ORM**; implemented an intelligent fallback mode using simulated responses to ensure system reliability when the AI API is unavailable.

AI-Powered Faculty Workload Management Portal — React.js | Node.js | MySQL | Cerberus API

- Automated duty assignment and substitution scheduling for **100+ faculty members** at Sathyabama Institute, reducing scheduling effort by **70%**; supported manual entry and **bulk CSV upload of 500+ records** per operation. GitHub: github.com/Soumya-Subhra-Datta/Smart_Faculty_Workload_management_Portal
- Built on **React.js** frontend, **Node.js** backend, and **MySQL** database; integrated **Cerberus API** for schema validation, reducing invalid entries by **80%** and cutting admin overhead **from hours to minutes**.

EDUCATION

Bachelor of Engineering (B.E.) — Computer Science and Engineering (Artificial Intelligence)

Sathyabama Institute of Science and Technology, Chennai, Tamil Nadu | 2023 – 2027 | CGPA: 7.85

CERTIFICATIONS

NPTEL — Machine Learning | IIT Kharagpur (2025)

NPTEL — Deep Learning | IIT Kharagpur (2026)

Here and Now AI — Generative AI & Agentic Architectures: Chatbots, RAG Systems and AI Workflow Automation (2025)

Here and Now AI — Full Stack Software Development using GitHub Speckit (2026)